Permeability from Sparse Data: Diffusion Results

A working permeability prior, and first conditioning on sensor data. Progress report on the latent-diffusion track.

What we built

The goal is to return not one permeability map but an ensemble consistent with the sparse sensor data, and to read its spread as the uncertainty. A diffusion model learns what permeability fields look like; an assimilation step steers them to match the measurements. The path below runs in 2D first, then in full 3D.

Focus : permeability only for now, as agreed. Porosity is deferred and easy to add back later.

Step 1. Compressing the field

A small autoencoder compresses each permeability map to a compact latent grid and rebuilds it. The diffusion runs in that compact space, and the assimilation adjusts a few thousand numbers per map instead of every cell, which is what keeps the whole loop affordable.

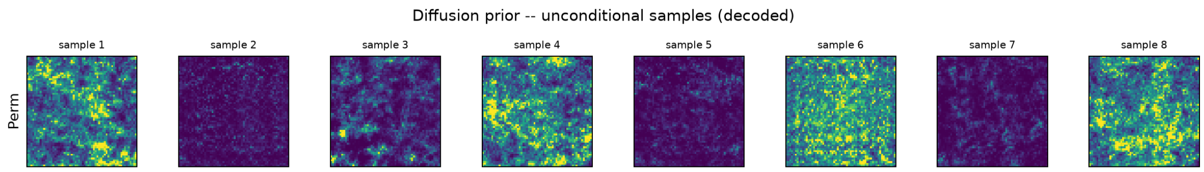
Field	Reconstruction R^2 (held-out)
Perm	0.974

The autoencoder rebuilds permeability at $R^2 = 0.974$, meaning it keeps almost all of the structure. That is a strong reconstruction and the standard bar for latent diffusion.

Step 2. A permeability generator

The diffusion model learns to turn noise into a realistic permeability field. Because it starts from noise, a different random start gives a different field, so sampling it repeatedly produces a whole ensemble with no extra machinery. The samples below are generated from scratch, not copied from the data.

Training : converged cleanly over 400 epochs to a denoising loss of 0.036.



Unconditional samples from the diffusion prior, decoded to permeability. The fields are varied and spatially structured, with high and low permeability zones, which is the raw material for the uncertainty spread.

Statistic	Generated	Real (train)
mean	-0.144	-0.000
std	1.107	1.000

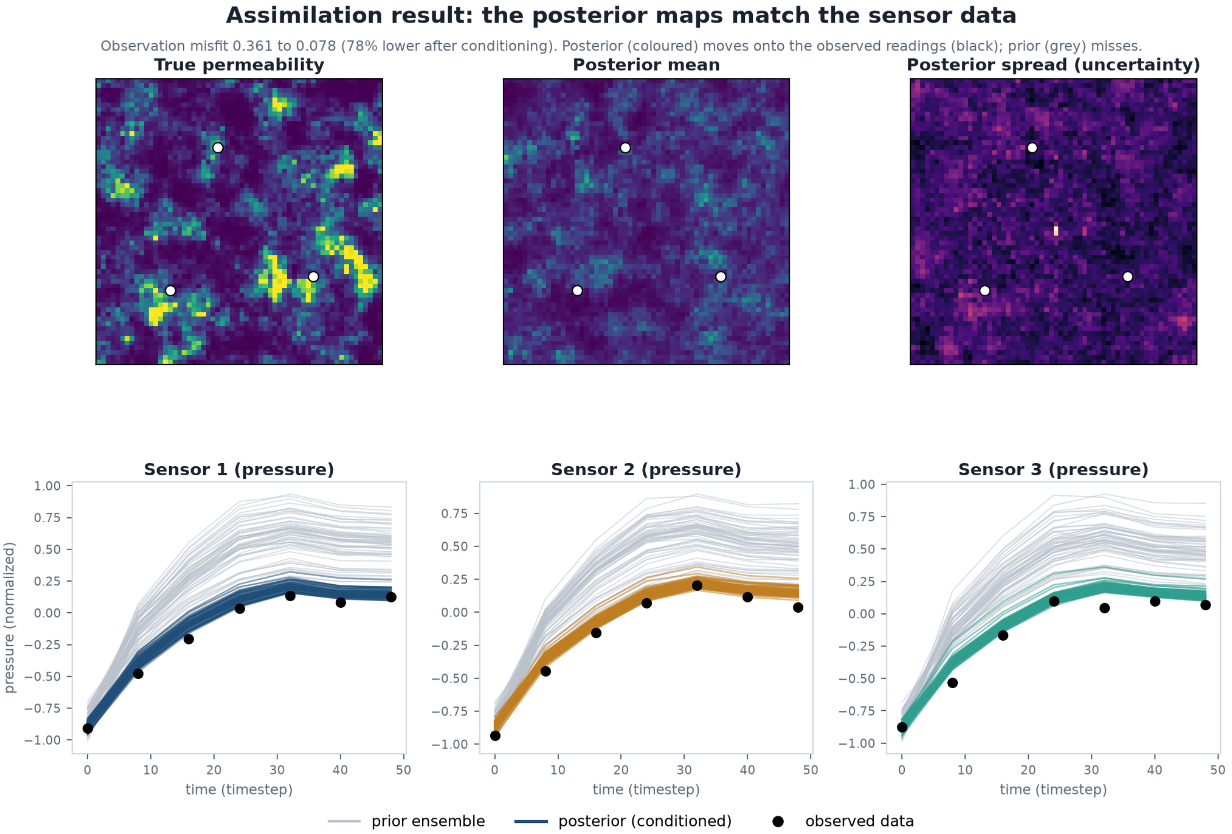
The generated fields sit close to the real data in mean and spread, so the prior is not just plausible by eye but statistically on target.

Step 3. Matching the sensor data

Assimilation is where the prior meets the measurements. We take the ensemble of candidate maps, predict what each one would read at the sensors using a fast forward surrogate, compare that to the observed readings, and nudge each map so its prediction moves toward the data. The check we can show is direct: the kept maps have to reproduce the readings at the right places and at the right times.

Quantity	Prior	Posterior
Observation misfit (RMSE)	0.361	0.078

Conditioning lowers the observation misfit by 78.5%: the ensemble moves from disagreeing with the sensors to matching them. The layout uses 60 members and 4 assimilation steps.



Assimilation result. Top: the true field, the posterior mean, and the posterior spread (the uncertainty), with sensors marked. Bottom: at each sensor, the posterior predictions (coloured) move onto the observed readings (black) while the prior (grey) misses. This is the data-driven version of the plan's Figure 4.

2D result: sensors match, field does not

In 2D the sensor signal is recovered but the permeability field is not: observation misfit drops 78.5%, yet the field R2 stays at -0.18 (negative = no better than the average field).

Metric	4 sensors (2D)	16 sensors + localization
Observation misfit reduction	78.5%	83.7%
Field R2 (posterior mean)	-0.18	-0.17
Near-sensor RMSE (lower is better)	1.40	1.28
Ensemble coverage of the truth	0.66	0.57

More sensors and localization barely help (R2 still -0.17): in 2D the readings depend little on any single permeability slice, so matching them says little about the rock. The bottleneck is the forward model, addressed next.

The fix: 3D volumes with the injection controls

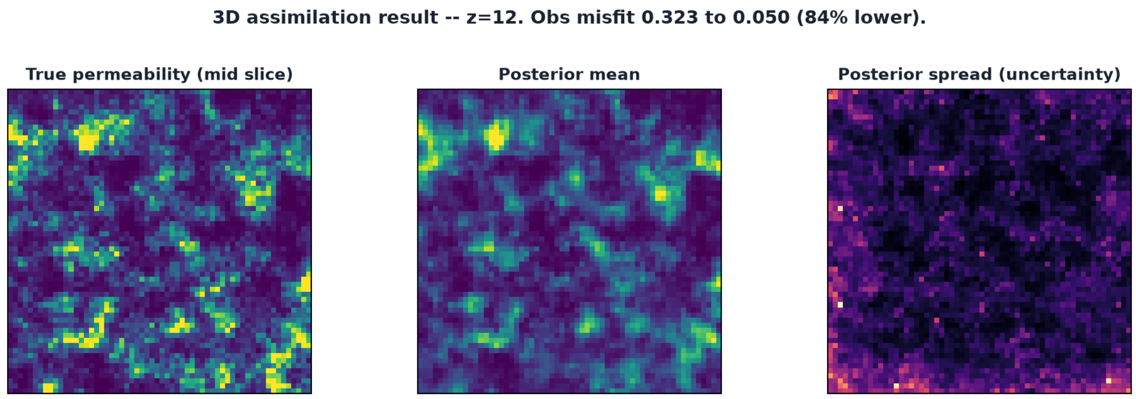
Two changes made the inverse work: full 3D volumes, and giving the forward model the injection program (rate, bottom-hole pressure, boundary), not just the rock. Pressure especially is driven by injection, and the dataset spans many injection scenarios, so a rock-only forward could not tell them apart.

Forward model accuracy (R2)	Perm-only (2D/3D)	With injection controls (3D)
Pressure	-0.22	0.89
Saturation	0.49	0.60

Pressure went from worse-than-guessing to well predicted once the forward could see the injection. An accurate forward is what lets the assimilation carry real information about the rock.

Field recovery	2D (best)	3D + controls
Perm R2 (posterior mean)	-0.17	0.79
Near-sensor RMSE (prior to post)	1.19 to 1.28	0.87 to 0.43
Ensemble coverage of the truth	0.57	0.84

With the accurate forward, the inverse now recovers the field. The permeability R2 goes from 0.22 to 0.79 (positive, and rising with conditioning, where 2D went negative), the near-sensor error drops, and the ensemble contains the true field in 84% of cells. Recovering a full field from a handful of sensors is not possible for any method, so the win is exactly this: the rock is recovered where the data constrains it, with a calibrated uncertainty band elsewhere.



3D assimilation (mid-depth slice): true permeability, posterior mean, and posterior spread. The posterior recovers the structure near the sensors; the spread marks what the sparse data leaves uncertain.

Summary: recovery vs. sensor count (3D)

Metric	8 sensors	48 sensors
Sensor layout	4 x,y x 2 depths	16 x,y x 3 depths
Observation misfit reduction	68%	84%
Perm R2 (prior to posterior)	0.16 to 0.37	0.22 to 0.79
Near-sensor RMSE (prior to posterior)	0.90 to 0.66	0.87 to 0.43
Ensemble coverage of the truth	0.89	0.84

More sensors sharpen the recovery (R2 0.37 to 0.79, near-sensor error roughly halved) while coverage stays well calibrated.